

The major networks that use machine learning

Traffic routing: One of the fundamental concepts essential for a network is routing, and this entails selecting a path for packet transmission. Routing takes into consideration cost minimisation, maximisation of link utilisation, operational policies and a few other attributes. Hence, ML models are challenged with the ability to cope and scale with today's dynamic and complex network topologies. They should also have the ability to learn the correlation between the selected path and then predict the consequences to be faced for a particular routing decision made. Reinforcement learning has done wonders in this aspect of traffic routing.

The initial use of reinforcement learning was done through the Q-routing (based on Q-learning) algorithm, in which a router 'X' learns to map a particular routing policy (for example, to destination 'D' via neighbour 'Y') to its Q-value. This Q-value is nothing but an estimate of the time that will be taken by the packet to reach 'D' via 'Y' including all the queue and transmission delays over the link.

Even though this Q-routing algorithm performs exceptionally well in a dynamically changing network topology, under heavy load the algorithm constantly changes the routing policy, which creates bottlenecks in the network. The most successful model was 'Team-Partitioned Opaque-Transition Reinforcement Learning (TPOT-RL)' proposed by Veloso and Stone. This algorithm has high computational complexity considering the very large number of states to be explored, and high communication overhead.

Traffic prediction: Network traffic prediction plays a major role in today's complex and diverse networks. Time series forecasting (TSF) is the major solution that helps forecast future traffic in a network. A TSF is a simple regression model that is capable

of drawing an accurate correlation between future traffic and previously observed traffic volumes.

The existing models for traffic prediction are statistical analysis models and supervised ML models. Statistical analysis models are usually built on the autoregressive integrated moving average (ARIMA) model, while the majority of learning is achieved via supervised neural networks. But due to the rapid growth of networks and the corresponding complexity of traffic, the traditional TSF models are compromised, which has led to the rise of advanced machine learning models.

As per the survey (<https://jisajournal.springeropen.com/articles/10.1186/s13174-018-0087-2#Sec1>) by Raouf Boutaba, "Eswaradass proposed an MLP-NN based bandwidth prediction system for grid environments and compared it to the Network Weather Service (NWS) bandwidth forecasting AR models for traffic monitoring and measurement. The goal of the system is to forecast the available bandwidth on a given path by feeding the NN with the minimum, maximum and average number of bits per second used on that path in the last epoch (ranging from 10s to 30s)."

Apart from the TSF based solutions, network traffic can also be predicted through non-TSF methods like Frequency Domain based methods in addition to Elephant flows for the network traffic flow. One of the non-TSF implementations incorporates the False Nearest Neighbour algorithm trained with backpropagation using simple gradient descent and wavelet transform to enable the model to capture both frequency and time features of the traffic time series.

Traffic classification: To perform a wide range of network operations, traffic classification is a must. This classification includes

capacity planning, security and intrusion detection, and performance monitoring. During an operation of a big network, unnecessary traffic in business-critical applications is a waste of resources.

We have predefined classes of networks such as HTTP, FTP, WWW, DNS, and P2P for training supervised models. But payload based traffic classification can be performed without any prior knowledge of the application classes using supervised learning, as shown by Ma *et al* in 'Unexpected Means of Protocol Inference' - 2006. For Host-Behaviour based traffic classification, the four features — service proximity, activity profiles, session duration and periodicity mentioned by Schatzmann — act as good discriminators for a support vector machine (SVM) classifier to distinguish between Web mail and non-Web mail traffic using a five-fold cross-validation scheme. One of the oldest and successful network classifiers was by Roughan, where he implemented a K-Nearest Neighbour and Linear Discriminant Analysis to map network traffic into different classes of interest.

Congestion control: The feature responsible for throttling the number of packets entering the network is congestion control. It also ensures network stability, fairness (resource utilisation) and packet loss ratio. A Bayesian packet loss classifier with up to 90 per cent detection probability on different data sets like BU and PMA, along with an analytic Markov Model (for evaluating TCP variant) enhanced with Bayesian packet loss classifier (by Fonseca and Crovella), resulted in a throughput improvement of up to 25 per cent on the classical TCP-Reno algorithm.

Coming to the optical network variants, the congestion control was tackled in optical burst switching (OBS) networks. The data was collected by simulation with OBS